

Official Exam Solutions: Jimma University Respa Microbiology Exam (2017)

Q1. Why is *Mycoplasma pneumoniae* resistant to penicillin?

- A. It produces high levels of beta-lactamase
- B. It has an impermeable outer membrane
- C. It has no cell wall
- D. It is an obligate intracellular parasite that replicates only inside host cells
- E. It has mutated penicillin-binding proteins (PBPs)
- **Correct Answer: C. It has no cell wall**
- **Deep Explanation:**
- ***M. pneumoniae* Biology:** As shown, *Mycoplasma pneumoniae* is a highly unique bacterial species because it "**has no cell wall**" and "**lacks peptidoglycan cell walls.**" Its cellular boundary is instead formed by a plastic, pleomorphic lipid membrane that contains sterols.
- **Mechanism of Penicillin Resistance:** Beta-lactam antibiotics (including penicillin, ampicillin, and cephalosporins) are cell wall synthesis inhibitors. They function by binding to Penicillin-Binding Proteins (PBPs) and disrupting the cross-linking of the peptidoglycan layer, which eventually leads to osmotic lysis of the bacterial cell.
- **Distractor Analysis:**
 - **A, B, and E** are incorrect because they represent acquired resistance mechanisms in species that *do* possess cell walls (such as *S. aureus* producing beta-lactamases or *S. pneumoniae* mutating PBPs). *Mycoplasma* exhibits inherent, natural resistance because there is no target (peptidoglycan) for the antibiotic to act on.
 - **D** is incorrect because *Mycoplasma pneumoniae* is an extracellular pathogen, not an obligate intracellular one. (Conversely, *Chlamydia pneumoniae* is an obligate intracellular pathogen).

Q2. How does the normal microbiota of the upper respiratory tract help in host defense mechanisms?

- A. By producing high levels of secretory IgA
- B. By competing for nutrients and attachment sites
- C. By directly phagocytosing invading pathogenic microbes
- D. By activating the classical complement pathway
- E. By lowering the viscosity of the respiratory mucus layer
- **Correct Answer: B. By competing for nutrients and attachment sites**
- **Deep Explanation:**
- **Role of Normal Flora:** The normal microbiota of the nasal cavity and throat represents a

crucial biological barrier to infection. It colonizes mucosal surfaces in healthy individuals, and its presence protects the host through competitive exclusion.

- **Mechanisms of Competition:** The resident flora is highly adapted to the mucosal environment and protects the host by:
 1. Consuming locally available carbohydrates, amino acids, and other trace **nutrients**, leaving incoming pathogens to starve.
 2. Physically occupying cell-surface receptors and **attachment sites** on epithelial cells, preventing transient pathogenic invaders from adhering and colonizing.
- **Distractor Analysis:**
 - **A and D** are host-mediated immunological defenses (secretory IgA is produced by host plasma cells in the mucosa, and complement proteins are synthesized by the liver) rather than functions of the bacterial microbiota.
 - **C** is a cellular host defense mediated by professional phagocytes (macrophages and neutrophils), not bacteria.
 - **E** is incorrect; lowering mucus viscosity is a virulence mechanism used by pathogens like the Influenza virus (via neuraminidase) to breach defenses, not a protective mechanism of the normal flora.

Q3. Which of the following is NOT considered part of the normal microbiota of the upper respiratory tract?

- **A. *Streptococcus pneumoniae***
- **B. *Neisseria meningitidis***
- **C. *Staphylococcus aureus***
- **D. *Mycobacterium tuberculosis***
- **E. *Haemophilus influenzae***
- **Correct Answer: D. *Mycobacterium tuberculosis***
- **Deep Explanation:**
- **Obligate Pathogen Status:** *Mycobacterium tuberculosis* is an obligate human pathogen that is never part of the normal, healthy resident flora. Its presence in clinical specimens always indicates either active tuberculosis disease or a latent tuberculosis infection.
- **The "Carrier Status" of URT:** The course materials explicitly note that the upper respiratory tract is continuously exposed to inhaled pathogens, and as a result, **"the normal microbiota of the nasal cavity and throat can include pathogenic microorganisms in a carrier status."** Healthy individuals commonly harbor potential pathogens such as *S. pneumoniae* (40–70% of healthy adults are carriers), *Staphylococcus aureus*, and *Haemophilus influenzae* as transient or resident commensals without active clinical symptoms.
- **Distractor Analysis: A, B, C, and E** are incorrect because they are all well-established members of the URT microbiota that can exist in healthy individuals in a carrier status.

Q4. Which of the following viruses can cause pharyngitis and tonsillitis?

- **A. Influenza virus**
- **B. Adenovirus**
- **C. Parainfluenza virus**
- **D. All of the above**
- **E. None of the above**
- **Correct Answer: D. All of the above**
- **Deep Explanation:**
- **Viral Etiology of URT Inflammation:** The course materials note that several families of viruses are common causes of upper respiratory tract infections, pharyngitis, and tonsillitis.
- **Pathogen Profiles:**
 - **Influenza virus** attaches and multiplies locally in the mucous membranes of the respiratory tract, causing rhinitis, pharyngitis, and bronchitis.
 - **Adenovirus** has multiple serotypes that cause upper respiratory infections, including rhinitis, pharyngitis, and tonsillitis.
 - **Parainfluenza virus** is a major cause of respiratory infections in infants and young children, causing croup and mild "common cold" symptoms such as pharyngitis.
- Therefore, since all three viruses are explicitly documented as causing pharyngitis and tonsillitis, **D** is the correct option.

Q5. What is the primary function of secretory IgA in host defense?

- **A. It activates the classical complement system to lyse foreign bacterial cells**
- **B. It acts as a direct cytotoxin against viral-infected epithelial cells**
- **C. It traps bacteria in mucin by attaching itself to the bacteria**
- **D. It stimulates mast cell degranulation and histamine release**
- **E. It degrades bacterial cell wall peptidoglycan**
- **Correct Answer: C. It traps bacteria in mucin by attaching itself to the bacteria**
- **Deep Explanation:**
- **Mucosal Immune Exclusion:** Secretory IgA (sIgA) is the principal immunoglobulin present in mucosal secretions (tears, saliva, colostrum, and respiratory mucus). Its primary protective mechanism is **immune exclusion**.
- **Trapping in Mucin:** Rather than acting as a direct cytolytic or a strong opsonin, sIgA binds to specific adhesive molecules (adhesins) on the surface of incoming bacteria. It cross-links the bacteria and anchors them to host mucin proteins in the respiratory tract. This physically traps the bacteria within the mucus blanket, preventing them from adhering to and invading the underlying epithelial cells. The trapped bacteria are then swept out of the airway by the ciliary escalator.
- **Distractor Analysis:**
 - **A** is incorrect because IgA is a poor activator of the classical complement pathway (which is primarily activated by IgM and IgG).
 - **B** is incorrect because IgA is non-cytotoxic; cellular cytotoxicity against virus-infected cells is mediated by CD8+ cytotoxic T lymphocytes and Natural Killer (NK) cells.
 - **D** is the primary function of IgE antibodies, not IgA.

- **E** is the enzymatic mechanism of lysozyme, which is a chemical barrier in secretions, not an antibody function.

Q6. Which component of *Streptococcus pneumoniae* is responsible for activating the classical complement system?

- **A.** C-polysaccharide
- **B.** M-protein
- **C.** Teichoic acid
- **D.** Pneumolysin
- **E.** Peptidoglycan
- **Correct Answer: D. Pneumolysin**
- **Deep Explanation:**
- **Pneumolysin Function:** As outlined, pneumolysin is an important virulence factor of *S. pneumoniae*. It is an intracellular cytolysin that is released when the pneumococcal cell undergoes autolysis.
- **Complement Activation:** Pneumolysin disrupts host cell membranes by forming transmembrane pores. Crucially, pneumolysin possesses structural homology to the Fc portion of IgG, which allows it to bind C1q directly and activate the classical complement cascade away from the bacterial cell surface. This depletes host complement proteins and triggers an intense, damaging localized inflammatory response in host tissues.
- **Distractor Analysis:**
 - **A and C** are structural cell wall components of *S. pneumoniae* but are not the primary activators of the classical complement system.
 - **B** (M-protein) is a major anti-phagocytic virulence factor of *Streptococcus pyogenes*, not *Streptococcus pneumoniae*.
 - **E** (peptidoglycan) is the structural mesh of the cell wall but does not directly drive classical complement activation in the same potent manner as pneumolysin.

Q7. Which component of the Influenza virus lowers the viscosity of mucus?

- **A.** Hemagglutinin
- **B.** Neuraminidase
- **C.** Matrix protein 1 (M1)
- **D.** Non-structural protein 1 (NS1)
- **E.** Nucleoprotein
- **Correct Answer: B. Neuraminidase**
- **Deep Explanation:**
- **Neuraminidase (NA) Enzymatic Action:** The envelope of the Influenza virus is studded with two major glycoprotein spikes: **Hemagglutinin (HA)** and **Neuraminidase (NA)**. NA is an enzyme (sialidase) that cleaves terminal sialic acid (neuraminic acid) residues from glycoproteins and glycolipids present in host secretions and on cell membranes.
- **Sialic Acid Cleavage:** By breaking down these sialic acid molecules, Neuraminidase

liquefies and **decreases the viscosity of the protective respiratory mucus blanket**. This allows the viral particles to penetrate the mucous barrier, reach the underlying epithelial cell receptors, and successfully establish infection.

- **Distractor Analysis:**

- **A** (Hemagglutinin) is responsible for **viral attachment** to sialic acid receptors on host cells, not for degrading mucus.
- **C** (M1 matrix protein) provides structural integrity to the viral envelope.
- **E** (Nucleoprotein) associates with viral RNA segments to form the internal helical nucleocapsid.

Q8. What are the most common causes of atypical pneumonia?

- **A.** *Streptococcus pneumoniae* and *Haemophilus influenzae*
- **B.** *Mycoplasma pneumoniae* and *Chlamydophila pneumoniae*
- **C.** *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*
- **D.** *Legionella pneumophila* and *Mycobacterium tuberculosis*
- **E.** *Pneumocystis jirovecii* and *Aspergillus fumigatus*
- **Correct Answer: B. *Mycoplasma pneumoniae* and *Chlamydophila pneumoniae***
- **Deep Explanation:**
- **Typical vs. Atypical Pneumonia:** "Typical" pneumonia (most commonly caused by *S. pneumoniae*) presents as acute, severe lobar consolidation with productive, purulent sputum. In contrast, "atypical" pneumonia presents with an insidious onset, dry, non-productive cough, low-grade fever, headache, and patchy interstitial infiltrates on chest X-ray.
- **Atypical Pathogens:** The two most common causes of atypical (often referred to as "walking") pneumonia are *Mycoplasma pneumoniae* and *Chlamydophila pneumoniae* (historically known as *Chlamydia pneumoniae*). These atypical pathogens do not grow on standard media and cannot be stained with Gram stain.
- **Distractor Analysis:**
 - **A** represents typical bacterial pathogens that cause classic lobar pneumonia.
 - **C** represents classic Gram-negative bacteria associated with severe, necrotizing hospital-acquired (nosocomial) pneumonia.
 - **D** includes *Legionella* (an atypical pathogen) and *Mycobacterium tuberculosis* (the classic agent of chronic cavitary tuberculosis).
 - **E** represents opportunistic fungal pathogens that cause severe pneumonia in immunocompromised individuals.

Q9. What is the initial therapy for an upper respiratory tract viral infection?

- **A.** Immediate administration of broad-spectrum antibiotics
- **B.** High-dose systemic corticosteroids
- **C.** Supportive and symptomatic care (e.g., hydration, analgesics, rest)
- **D.** Immediate antiviral therapy with oseltamivir

- E. Inhaled bronchodilators and oxygen therapy
- **Correct Answer: C. Supportive and symptomatic care (e.g., hydration, analgesics, rest)**
- **Deep Explanation:**
- **Self-Limiting Nature:** The course materials highlight that most respiratory infections are **self-limiting**, meaning they resolve spontaneously as the host's immune system clears the viral pathogen.
- **Therapeutic Approach:** Because there are no routinely indicated antiviral drugs for uncomplicated common colds (caused by rhinoviruses, adenoviruses, or coronaviruses), and because antibiotics are entirely ineffective against viruses, the standard first-line clinical management consists of **supportive and symptomatic care** (rest, hydration, and analgesics like acetaminophen to relieve fever and pain).
- **Distractor Analysis:**
 - **A** (broad-spectrum antibiotics) is clinically inappropriate and violates antimicrobial stewardship; antibiotics are indicated only when a secondary bacterial infection (such as bacterial otitis media or sinusitis) is confirmed.
 - **B** (corticosteroids) is not indicated for routine viral infections and can suppress necessary antiviral immune responses.
 - **D** (oseltamivir) is a specific neuraminidase inhibitor indicated only for influenza infections, and is not used for general URT viral infections (like rhinoviruses or adenoviruses).
 - **E** is reserved for severe lower respiratory distress (such as severe pneumonia or asthma exacerbations), not routine URT infections.

Q10. What is the first-line therapy for a new case of Tuberculosis (TB)?

- A. Penicillin, Erythromycin, and Amoxicillin
- B. Isoniazid, Rifampin, Pyrazinamide, and Ethambutol
- C. Streptomycin, Clarithromycin, and Levofloxacin
- D. Vancomycin, Linezolid, and Doxycycline
- E. Ciprofloxacin, Metronidazole, and Azithromycin
- **Correct Answer: B. Isoniazid, Rifampin, Pyrazinamide, and Ethambutol**
- **Deep Explanation:**
- **The RIPE Regimen:** As detailed, effective treatment of active, drug-susceptible tuberculosis requires a combination of multiple first-line bactericidal and bacteriostatic drugs for a duration of 6 to 9 months. The standard initial phase (intensive phase) consists of a four-drug regimen: Isoniazid (INH/H), Rifampin (RIF/R), Pyrazinamide (PZA/Z), and Ethambutol (EMB/E).
- **Rationale for Combination Therapy:** Using four distinct drugs with different mechanisms of action simultaneously ensures that rapidly dividing extracellular bacilli are killed (INH, RIF), slow-growing intracellular bacilli within macrophages are targeted (PZA), and the selection of drug-resistant mutants is prevented.

- **Distractor Analysis:**
 - **A** contains standard beta-lactams and macrolides that have zero activity against the thick, waxy, mycolic acid-containing cell wall of *M. tuberculosis*.
 - **C** contains streptomycin (a first-line drug) along with levofloxacin and clarithromycin, which are used in second-line or alternative regimens when resistance to first-line agents is present.
 - **D and E** are standard antibacterial regimens for typical/atypical bacterial infections but have no role in first-line tuberculosis therapy.

Q11. What is Tuberculosis (TB) that is resistant to at least rifampin and isoniazid called?

- **A.** Extensively drug-resistant TB (XDR-TB)
- **B.** Multi-drug resistant TB (MDR-TB)
- **C.** Totally drug-resistant TB (TDR-TB)
- **D.** Primary resistant TB
- **E.** Non-reactive TB
- **Correct Answer: B. Multi-drug resistant TB (MDR-TB)**
- **Deep Explanation:**
- **MDR-TB Definition:** The course materials explicitly define **Multidrug resistance (MDR)** as resistance to "**at least two of the best anti-TB drugs, isoniazid and rifampicin.**" These are the two most powerful first-line drugs used to treat TB, and resistance to both significantly complicates therapy, requiring longer, more expensive, and more toxic second-line regimens.
- **Distractor Analysis:**
 - **A** (Extensively Drug-Resistant, XDR-TB) is defined as resistance to isoniazid and rifampin **PLUS** resistance to any fluoroquinolone and at least one of the three injectable second-line drugs (Amikacin, Kanamycin, or Capreomycin).
 - **C** (Totally Drug-Resistant) is a term used for strains resistant to all tested first-line and second-line drugs, which is not the specific definition of resistance to isoniazid and rifampin alone.
 - **D** (Primary resistance) refers to contracting a drug-resistant strain from another individual rather than developing resistance during treatment.
 - **E** is not a drug resistance classification.

Q12. Which of the following is a common cause of hospital-acquired pneumonia?

- **A.** *Mycoplasma pneumoniae*
- **B.** *Pseudomonas aeruginosa*
- **C.** *Streptococcus pyogenes*
- **D.** *Bordetella pertussis*
- **E.** *Corynebacterium diphtheriae*
- **Correct Answer: B. Pseudomonas aeruginosa**

- **Deep Explanation:**
- **Hospital-Acquired (Nosocomial) Pneumonia:** Nosocomial pneumonia typically develops in hospitalized patients (especially those on mechanical ventilators) who are exposed to invasive devices, broad-spectrum antibiotics, and resistant hospital flora.
- **Etiology of Nosocomial Infections:** The course materials list the primary bacterial pathogens associated with hospital-acquired pneumonia: **Enterobacteriaceae**, ***Pseudomonas aeruginosa***, and ***Staphylococcus aureus***. These Gram-negative opportunistic bacilli are highly pathogenic and often possess multi-drug resistance.
- **Distractor Analysis:**
 - **A** (*Mycoplasma pneumoniae*) is the classic cause of community-acquired atypical pneumonia in young, healthy populations, not nosocomial environments.
 - **C** (*Streptococcus pyogenes*) can cause severe empyema but is not a primary, common cause of hospital-acquired pneumonia.
 - **D and E** (*B. pertussis* and *C. diphtheriae*) are highly contagious pediatric pathogens transmitted via respiratory droplets, causing pertussis (whooping cough) and diphtheria respectively, rather than opportunistic hospital-acquired pneumonias.

Q13. Considering a 48-year-old patient presenting with fever, dry cough, and loss of taste and smell, what is the most likely causative agent?

- **A.** Influenza A virus
- **B.** Respiratory Syncytial Virus (RSV)
- **C.** *Mycoplasma pneumoniae*
- **D.** SARS-CoV-2 (COVID-19)
- **E.** *Streptococcus pneumoniae*
- **Correct Answer: D. SARS-CoV-2 (COVID-19)**
- **Deep Explanation:**
- **Clinical Presentation:** Fever and dry cough are non-specific symptoms shared by many viral and atypical respiratory pathogens. However, a sudden loss of taste (ageusia) and loss of smell (anosmia) is a highly specific, pathognomonic clinical feature of infection with **SARS-CoV-2 (COVID-19)**.
- **Pathobiology of Anosmia:** SARS-CoV-2 infects and damages the sustentacular support cells of the olfactory epithelium in the nasal cavity, which leads to sudden, temporary olfactory dysfunction that is not seen in typical influenza or bacterial pneumonias.
- **Distractor Analysis:**
 - **A and B** (Influenza and RSV) cause severe upper and lower respiratory symptoms but do not characteristically present with acute, complete loss of taste and smell.
 - **C** (*Mycoplasma pneumoniae*) causes atypical pneumonia with a persistent dry cough but lacks these neuropathic sensory symptoms.
 - **E** (*Streptococcus pneumoniae*) causes typical lobar pneumonia, characterized by high fever and productive cough with purulent or "rusty" sputum.

Q14. What is the primary function of secretory IgA in the respiratory mucosa wall?

- A. It activates the alternative complement pathway
- B. It degrades bacterial peptidoglycan
- C. It traps bacteria in mucin by attaching itself to the bacteria
- D. It stimulates mast cell degranulation
- E. It acts as a direct cytotoxin against viral-infected cells
- **Correct Answer: C. It traps bacteria in mucin by attaching itself to the bacteria**
- **Deep Explanation:**
- **Mechanics of sIgA:** Upon secretion onto the mucosal surfaces of the respiratory mucosa wall, secretory IgA (sIgA) functions as an immunological shield. It binds directly to specific adhesive antigens on the surface of invading pathogens, neutralizing them through immune exclusion. By attaching itself to both the pathogen and the surrounding host mucin fibers, it physically immobilizes the bacteria, preventing them from accessing and adhering to underlying epithelial cell receptors.
- **Complementary to Q5:** This question reinforces the clinical importance of sIgA in the mucosal lining.
- **Distractor Analysis:**
 - A is incorrect because IgA does not efficiently activate either complement pathway on its own.
 - B is the function of Lysozyme, which is a chemical enzyme, not an antibody.
 - D is mediated by IgE antibodies binding to IgE Fc receptors on mast cells.
 - E is incorrect because antibodies cannot directly lysing host cells without complement or effector cell engagement.

Q15. Which of the following are opportunistic fungi that commonly emerge due to the prolonged use of mechanical ventilation or antibiotics?

- A. *Cryptococcus neoformans* and *Histoplasma capsulatum*
- B. *Candida* species and *Aspergillus* species
- C. *Coccidioides immitis* and *Blastomyces dermatitidis*
- D. *Saccharomyces cerevisiae* and *Pneumocystis jirovecii*
- E. *Sporothrix schenckii* and *Microsporum canis*
- **Correct Answer: B. *Candida* species and *Aspergillus* species**
- **Deep Explanation:**
- **Opportunistic Fungal Risk Factors:** Prolonged use of broad-spectrum antibiotics decimates the host's competitive resident bacterial flora, while mechanical ventilation breaches the primary mechanical defenses of the upper respiratory tract. This severe disruption of host defenses permits the rapid overgrowth of opportunistic fungi.
- **Primary Opportunistic Molds & Yeasts:** *Aspergillus* species (a filamentous mold) and *Candida* species (an opportunistic yeast) are the most common hospital-acquired fungal

pathogens that colonize and invade tissue under these clinical conditions.

- **Distractor Analysis:**

- **A, C, and D** include endemic or specific environmental fungi (*Cryptococcus*, *Histoplasma*, *Coccidioides*, *Blastomyces*) that are acquired via inhalation in specific geographic regions and cause systemic mycoses, rather than being common opportunistic colonizers directly associated with broad-spectrum antibiotic use in hospitalized patients.
- **E** represents *Sporothrix* (which causes subcutaneous rose-gardener's disease) and *Microsporum* (a dermatophyte causing cutaneous ringworm), neither of which are respiratory pathogens.

Q16. Which fungal agent typically affects and causes severe opportunistic infections in AIDS patients?

- **A. *Pneumocystis jirovecii***
- **B. *Trichophyton rubrum***
- **C. *Sporothrix schenckii***
- **D. *Microsporum canis***
- **E. *Saccharomyces cerevisiae***
- **Correct Answer: A. *Pneumocystis jirovecii***
- **Deep Explanation:**
- ***P. jirovecii* Pathobiology:** As noted in the course materials, *Pneumocystis jirovecii* (formerly *P. carinii*) is an extracellular fungus whose growth is restricted to the surfactant layer above the alveolar epithelium. It is a harmless commensal in healthy lungs but acts as a highly lethal opportunistic pathogen in patients with severe cell-mediated immunodeficiency, serving as a classic **AIDS-defining illness**.
- **Interstitial Plasma Cell Pneumonitis:** Uncontrolled multiplication of the fungus causes the alveoli to fill with a thick, foamy exudate, leading to severe hypoxia and interstitial pneumonitis.
- **Distractor Analysis:**
 - **B and D** are superficial dermatophytes that cause tinea infections of the skin, nails, and hair, rather than life-threatening systemic opportunistic pneumonias.
 - **C** is the causative agent of sporotrichosis, a subcutaneous fungal infection.
 - **E** is baker's yeast, which is generally non-pathogenic.

Q17. Which of the following is NOT normal flora of the upper respiratory tract (URT)?

- **A. *Staphylococcus***
- **B. *Neisseria meningitidis***
- **C. *Alpha Streptococcus***
- **D. *Mycobacterium tuberculosis***
- **E. *Streptococcus mutans***
- **Correct Answer: D. *Mycobacterium tuberculosis***

- **Deep Explanation:**
- **Obligate Pathogenicity:** *Mycobacterium tuberculosis* is an obligate pathogen with highly developed virulence mechanisms designed to survive and replicate inside alveolar macrophages. It is never part of the normal commensal flora of healthy individuals.
- **Resident URT Flora:** The upper respiratory tract is heavily colonized.
 - **A and B** are common colonizers of the nose and pharynx.
 - **C** represents alpha-hemolytic streptococci (such as *S. pneumoniae* or the *Viridans* group), which are dominant resident flora of the throat.
 - **E** (*Streptococcus mutans*) is a member of the *Viridans* group that resides in the oral cavity and is heavily involved in dental caries.
- Thus, **D** is the only correct answer.

Q18. Which of the following diseases/pathogens is characteristically manifested by tonsillar exudates and cervical lymphadenopathy (LAP)?

- **A. *Corynebacterium diphtheriae***
- **B. *Bordetella pertussis***
- **C. *Streptococcus pyogenes***
- **D. *Mycobacterium tuberculosis***
- **E. *Mycoplasma pneumoniae***
- **Correct Answer: C. *Streptococcus pyogenes***
- **Deep Explanation:**
- **Streptococcal Pharyngitis/Tonsillitis:** Acute pharyngitis and tonsillitis caused by *Streptococcus pyogenes* (Group A Strep) characteristically present with rapid-onset sore throat, high fever, painfully swollen tonsils covered in thick, **purulent tonsillar exudates**, and tender, enlarged **cervical lymph nodes (lymphadenopathy/LAP)**.
- **Distractor Analysis:**
 - **A** (*C. diphtheriae*) presents with a tough, greyish-white **pseudomembrane** rather than a loose tonsillar exudate, and is characterized by "bull neck" soft tissue swelling rather than simple localized lymphadenopathy.
 - **B** (*B. pertussis*) infects the ciliated epithelial cells of the trachea and bronchi, presenting with severe paroxysmal coughing fits without tonsillar exudates.
 - **D and E** (*M. tuberculosis* and *M. pneumoniae*) cause lower respiratory infections (tuberculosis and walking pneumonia) rather than acute, exudative tonsillopharyngitis.

Q19. Which of the following is the causative agent described as causing fever, cough, fatigue, loss of taste or smell, shortness of breath, and potentially epiglottitis?

- **A. *Haemophilus influenzae* type b**
- **B. *Streptococcus pneumoniae***
- **C. COVID-19**

- **D. *Bordetella pertussis***
- **E. *Corynebacterium diphtheriae***
- **Correct Answer: C. COVID-19**
- **Deep Explanation:**
- **Pathognomonic Constellation:** While fever, cough, fatigue, and shortness of breath are common to many respiratory pathogens, the addition of **loss of taste or smell** is highly specific to **COVID-19 (SARS-CoV-2)**. In addition, severe cases of COVID-19 can rarely present with upper airway swelling and epiglottitis-like complications.
- **Distractor Analysis:**
 - **A** (*Haemophilus influenzae* type b) is the classic cause of bacterial epiglottitis in children but does not present with olfactory or gustatory dysfunction.
 - **B and D** do not cause loss of taste and smell.
 - **E** (*C. diphtheriae*) causes severe pharyngeal pseudomembranes but is not associated with sensory loss.

Q20. Which component of the Influenza virus decreases the viscosity of host mucus?

- **A. Hemagglutinin (HA)**
- **B. Neuraminidase (NA)**
- **C. Matrix protein 1 (M1)**
- **D. Non-structural protein 1 (NS1)**
- **E. Nucleoprotein (NP)**
- **Correct Answer: B. Neuraminidase (NA)**
- **Deep Explanation:**
- **Mucus Degradation by NA:** This question is identical in concept to Q7. Neuraminidase (NA) is a key envelope glycoprotein spike that acts as a sialidase enzyme. By cleaving sialic acid residues on host glycoproteins and mucins, NA liquefies and **decreases the viscosity of the respiratory mucus layer**, enabling the virions to traverse the mucus barrier and infect host epithelial cells.
- **Distractor Analysis:** Refer to Q7 for the detailed structural and functional distractor analysis of HA, M1, and NP.

Q21. A patient presents with "bull neck" lymphadenopathy. Which of the following is the classic bacterial cause associated with this clinical sign and/or bacterial epiglottitis?

- **A. *Corynebacterium diphtheriae*** (classic cause of "bull neck" lymphadenopathy)
- **B. *Haemophilus influenzae*** (classic cause of bacterial epiglottitis)
- **C. *Bordetella pertussis***
- **D. *Mycoplasma pneumoniae***
- **E. Both A and B are correct descriptions for these respective clinical signs**
- **Correct Answer: E. Both A and B are correct descriptions for these respective clinical signs**

- **Deep Explanation:**
- **Pairing Pathogens with Clinical Signs:**
 - ***Corynebacterium diphtheriae*:** Is the classic cause of "**bull neck**" diphtheria. This sign is caused by severe swelling of the cervical lymph nodes accompanied by extensive soft-tissue edema of the neck, which is a major diagnostic indicator of pharyngeal diphtheria.
 - ***Haemophilus influenzae*:** Specifically, encapsulated **type b (Hib)**, is the classic cause of **acute bacterial epiglottitis**. This is a sudden, life-threatening inflammation of the epiglottis that presents with high fever, neck pain, inspiratory stridor, and a risk of fatal airway obstruction.
- Therefore, since both statements A and B are accurate, **E** is the correct answer.

Q22. Exposure to sick cattle is a classic risk factor for contracting which of the following diseases?

- **A. Brucellosis**
- **B. Q fever**
- **C. Anthrax**
- **D. Tuberculosis**
- **E. Leptospirosis**
- **Correct Answer: C. Anthrax**
- **Deep Explanation:**
- **Zoonotic Transmission of Anthrax:** *Bacillus anthracis* is primarily a pathogen of herbivores (goats, sheep, **cattle**, and horses). Humans acquire the infection through accidental contact with infected animals or their contaminated products (hides, hair, wool, or carcass fluids). Spores can enter through cuts in the skin (cutaneous anthrax) or be inhaled (pulmonary anthrax, classically known as "wool sorter's disease"). Therefore, exposure to sick cattle is a classic risk factor for contracting anthrax.
- **Note on Tuberculosis:** While *Mycobacterium bovis* can be transmitted via the ingestion of contaminated milk from infected cattle, causing bovine tuberculosis in humans, the direct occupational exposure to sick cattle carcasses and wool is more classically associated with *Bacillus anthracis* (anthrax) in general microbiology curricula.

Q23. A patient presenting with a paroxysmal cough followed by an inspiratory "whoop" is most likely infected with which pathogen?

- **A. *Streptococcus pneumoniae***
- **B. *Bordetella pertussis***
- **C. *Corynebacterium diphtheriae***
- **D. Influenza virus**
- **E. *Mycoplasma pneumoniae***
- **Correct Answer: B. *Bordetella pertussis***
- **Deep Explanation:**
- ***B. pertussis* Pathogenesis:** *Bordetella pertussis* is a small, encapsulated Gram-negative

coccobacillus that causes pertussis, or whooping cough. The bacteria specifically colonize the ciliated epithelial cells of the respiratory mucosa and release toxins that damage cells and paralyze the ciliary escalator.

- **The "Whoop" Sign:** Following a mild catarrhal stage, the disease progresses to the classic **paroxysmal stage**. This is characterized by repetitive, explosive coughing fits (paroxysms) during which the patient cannot inhale, ending with a sudden, forced inspiratory gasp against a partially closed glottis that produces the classic high-pitched "**whoop**".
- **Distractor Analysis: A, C, D, and E** are incorrect because they cause pneumonias, pseudomembrane obstruction, or flu-like symptoms, but do not produce the classic paroxysmal coughing paroxysms and high-pitched whoop characteristic of pertussis.

Q24. A patient who recently returned from a trip complains of a complete loss of taste and smell. What is the most likely cause?

- **A. Rhinovirus**
- **B. Adenovirus**
- **C. Coronavirus (COVID-19)**
- **D. Influenza virus**
- **E. Respiratory Syncytial Virus (RSV)**
- **Correct Answer: C. Coronavirus (COVID-19)**
- **Deep Explanation:**
- **Diagnostic Significance:** As established in Q13, sudden and complete loss of taste (ageusia) and loss of smell (anosmia) is a highly specific, pathognomonic clinical indicator of infection with **SARS-CoV-2, which is a member of the Coronavirus family**.
- **Distractor Analysis:** While rhinoviruses, adenoviruses, influenza, and RSV cause typical upper respiratory cold and flu symptoms (such as congestion, sore throat, or cough), they do not cause the acute, complete olfactory and gustatory sensory loss characteristic of COVID-19.